

Advanced Machine Learning

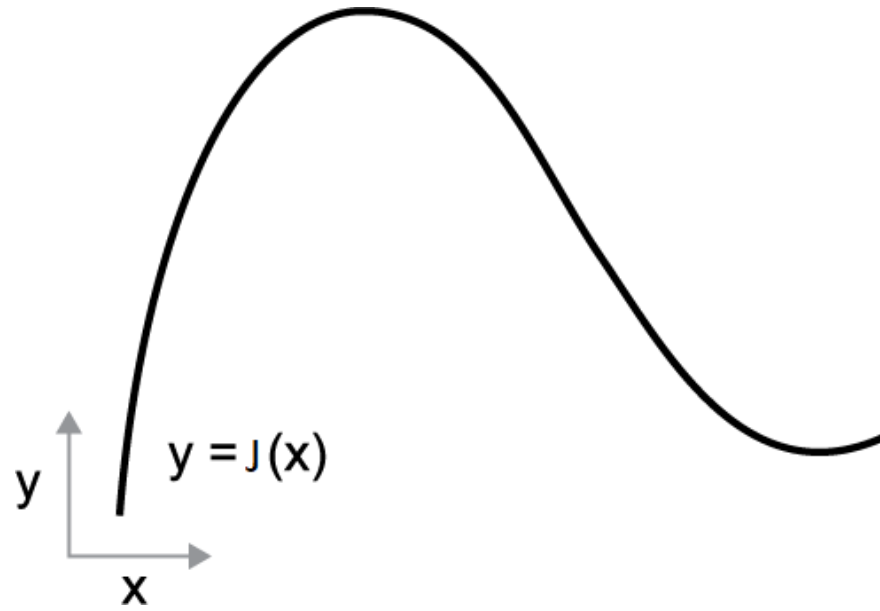
DATA 642

Lecture 1 — Introduction to derivatives

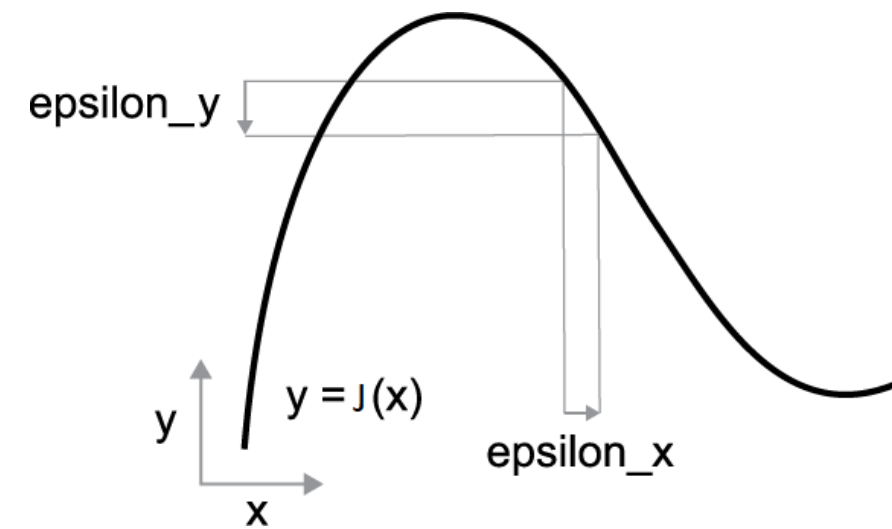
*Figures and notes are from the books Deep Learning with Python, Second Edition by François Chollet, Manning, ISBN 9781617296864 and Hands-On Machine Learning with Scikit-Learn Keras and TensorFlow, Aurelien Geron, O'Reilly

What is a derivative?

Let us consider a smooth function $J(x) = y$ that maps a number, x , to a new number y .



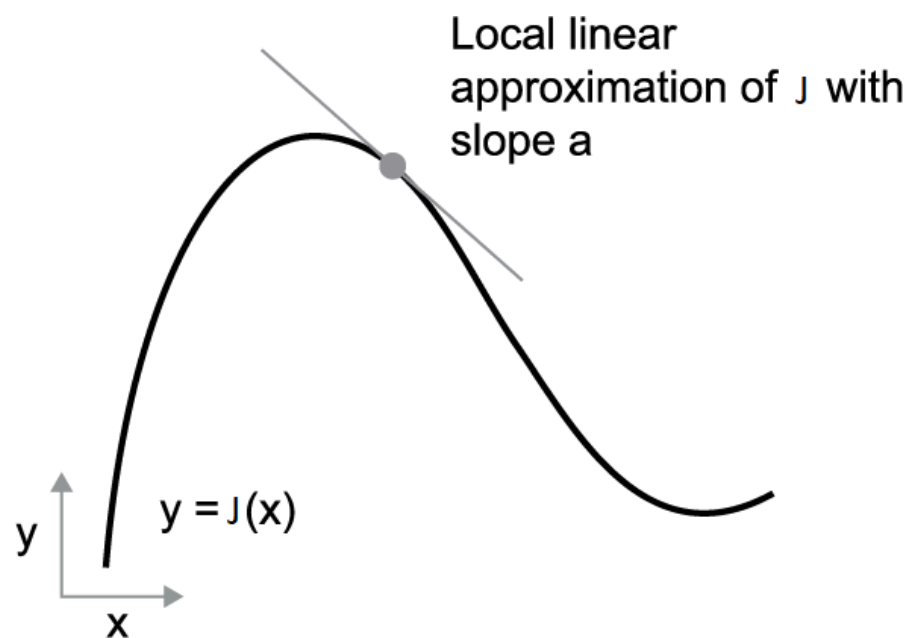
Because the function is **continuous**, a small change in x can only result in a small change in y . This is the intuition behind continuity. Let us say you increase x by a small factor, ϵ_x : this results in a small ϵ_y change to y .



In addition, because the function is smooth (its curve does not have any abrupt angles), when ϵ_x is small enough, around a certain point p , it is possible to approximate J as a linear function of slope a , so that ϵ_y becomes $a\epsilon_x$.

The slope a is called the derivative of J in p .

- If a is negative, it means a small increase in x around p will result in a decrease of $J(x)$.
- If a is positive, a small increase in x will result in an increase of $J(x)$.
- The absolute value of a (the **magnitude** of the derivative) tells you how quickly this increase or decrease will happen.



Being able to differentiate functions is a very powerful tool when it comes to optimization, the task of finding values of x that minimize the value of $J(x)$

- The function we were just looking at turned a scalar value x into another scalar value y : you could plot it as a curve in a 2D plane.
- Now imagine a function that turns a tuple of scalars (x, y) into a scalar value z . You could plot it as a 2D surface in a 3D space indexed by coordinates x, y, z .
- The concept of derivative can be applied to any such function, as long as the surfaces they describe are continuous and smooth. In mathematics when we go into higher dimensions we use the notion of **gradient**.